Ioanna Garefalaki

Curriculum Vitae

Evolutionary Biology Division
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Research Interests

Studied biology and learned how to code to make sense of it. I investigate how (epi)genetic variation arises, is regulated, and shapes the phenotypic traits that make life forms so remarkable.

Education

2023–present **PhD, Avian Epigenetics & Quantitative Genetics**, *Evolutionary Biology Division*, Ludwig-Maximilians-Universität Munich, Germany, International Max Planck Research School for Biological Intelligence.

2017 – 2019 **MSc, Molecular Biology & Plant Biotechnology**, *Faculty of Biology*, University of Crete, Heraklion, Greece.

2013 – 2017 **BSc, Biology**, *Biology Department*, University of Patras, Patra, Greece.

2009 – 2013 **BSc, Biomedical Science (Technological Education)**, *Biomedical Laboratories department*, University of West Attica, Athens, Greece.

Publications

Peer-Reviewed Journal Articles

2026 **Garefalaki, I.**, Mueller, S., Metzler, D., et al. "Technical Considerations in DNA Methylation Studies in Natural Populations." *In preparation for submission*.

2026 **Garefalaki, I.**, Wolf, J. B. W., et al. "Investigating the Impact of Epigenetic Variation on Evolution in Wild *Cyanistes caeruleus* Pedigrees." *In preparation*.

2026 Trougkakos, I., ... **Garefalaki, I.**, ... Terpos, E. "Integrated Transcriptomic and Proteomic Signatures of the BNT162b2 mRNA Vaccine in Human Blood." *Manuscript in resubmission*.

2021 Vasilarou, M., Alachiotis, N., **Garefalaki, I.**, Beloukas, A., & Pavlidis, P. "Population Genomics Insights into the First Wave of COVID-19." *Life*, 11(2), 129. doi: 10.3390/life11020129.

In Conference Proceedings

2026 **Garefalaki, I.**, Mueller, S., et al. "A statistical framework for DNA methylation repeatability in natural populations." *Annual Meeting of the Society for Molecular Biology & Evolution (SMBE)*, Copenhagen, Denmark. (Submitted: Oral Presentation)

2026 **Garefalaki, I.**, Mueller, S., et al. "Technical repeatability of DNA methylation in two wild avian species." *Evolutionary Genetics Community Conference (EvoGenMunich)*, Munich, Germany. (Submitted: Oral Presentation)

2025 **Garefalaki, I.**, Mueller, S., et al. "Investigating the Impact of Epigenetic Variation on Evolution in Wild *Cyanistes caeruleus* Pedigrees." *16th Int. Congress on the Zoogeography and Ecology of Greece and Adjacent Regions (ICZEGAR)*, Athens, Greece. (Poster)

2019 **Garefalaki, I.** & Pavlidis, P. "Evolutionary Forces in the Genomic Neighborhoods of Polymorphic Transposable Elements in Plant Populations." *12th FORTH Scientific Retreat*, Patras, Greece. (Poster)

2018 **Garefalaki, I.**, Chorba, K., & Mitsainas, G. P. "The Traits of Contact Zones between Different Chromosomal Races of *Mus musculus domesticus* in Two Robertsonian Systems of Greece." *6th Int. Conference of Rodent Biology and Management*, Potsdam, Germany. (Poster)

Research Experience

Ludwig Maximilian University Munich (LMU), Germany

Advisor **Prof. Jochen B. W. Wolf**, Chair of Evolutionary Biology; Max Planck Fellow, MPI for Biological Intelligence

2026 – Present ***Quantitative Genetic Modeling of Epigenetic Heritability***.
Implementing Linear Mixed Models (LMMs) to estimate 5mC heritability within a large-scale natural pedigree of 1200+ individuals. Validating model performance against morphological benchmarks and investigating heritable variation within CpG islands.

2023 – 2026 ***Methodological Optimization for Wild Population Epigenomics***.
Developed a robust statistical framework to assess CpG site repeatability and technical noise in RRBS/EM-seq datasets from natural populations using 372 technical replicates.

2023 – 2025 ***Strategic Coordination & Bioinformatic Optimization of EM-seq Data***.
Led the technical and financial strategy for a **€130k+ sequencing project** involving 1,288 libraries.
- **Project Management:** Performed independent read-depth modeling to achieve 30x target coverage; evaluated cost-coverage trade-offs across NovaSeq 6000 and NovaSeq X platforms to optimize DFG-funded resource allocation.

University of Thessaly, Greece

Advisor **Prof. Artemis G. Hatzigeorgiou**, Department of Informatics and Telecommunication

2022 ***Deep Learning Evaluation for microRNA Target Prediction***.
Developed standardized bioinformatics pipelines for comparison of *microT-CNN* against state-of-the-art target prediction tools to evaluate MRE counts and miRNA interactions accurancy.

BSRC "Alexander Fleming", Greece

Advisor **Dr. Christoforos Nikolaou**, Computational Genomics Group

2021 – 2022 ***Multi-omic Profiling of Post-Vaccination Immune Responses***.
Executed differential expression and multivariate analysis on high-dimensional whole-blood proteomic and transcriptomic datasets. Investigated protein correlation structures and pathway enrichment across healthy, elderly, and Multiple Myeloma cohorts to identify cohort-specific immune signatures.

Foundation for Research and Technology Hellas (FORTH), Greece

2018 – 2019 ***Evolutionary Dynamics of Transposable Elements (TEs) in Plants***.
Developed an open-source pipeline to detect putative adaptive signals and selective sweeps in genomic regions flanking polymorphic TE insertions in *Oryza sativa*.
Advisor: Prof. Pavlos Pavlidis, Bioinformatics and Population Genomics Group

2018 ***Functional Characterization of Xylella Virulence Factors***.
Utilized molecular cloning and transient expression assays to investigate pathogenic effectors within plant–pathogen interaction systems.
Advisor: Prof. Panagiotis Sarris, IMBB, Microbial Pathogenicity Laboratory

2018 ***Phylogenetic Analysis of Plant NLR Proteins***.
Comparative evolutionary analysis of nucleotide-binding leucine-rich repeat immune receptors in plants.
Advisor: Prof. Pavlos Pavlidis, Bioinformatics and Population Genomics Group

2018 ***CRISPR/Cas9 Screening***.
Performed molecular validation of DCL knock-out lines in *N. benthamiana* and conducted comparative evolutionary analysis of plant immune receptors.
Advisor: Prof. K. Kalantidis, Molecular Plant Biology Laboratory

University of Patras, Greece

Advisor **Dr. George P. Mitsainas**, Laboratory of Cytogenetics

2016 – 2017 ***Chromosomal Diversity in Mus musculus domesticus***.
Analyzed hybrid zones between Robertsonian races and acrocentric populations; documented complex chromosomal variation through karyotype analysis.

Fellowships & Awards

2020 ***2nd Award*** for best presentation at the 8Th Panhellenic Meeting of AIDS and Hepatitis 2020 "Population Genomics Insights into the First Wave of COVID-19".

Academic Achievements & Recognitions

2023 Selected for the **International Max Planck Research School – Biological Intelligence (IMPRS-BI)**, a competitive doctoral program of Max Planck Institute.

2013 Admitted to the Department of Biology, University of Patras, through national competitive transfer examinations (*Katataktiries Exetaseis*) with an **Excellent** grade.

Computational & Bioinformatics Skills

Programming R (*tidyverse*, *data.table*), Python, Bash/Shell, LaTeX; Git-based version control and automated reporting.

NGS & Epigenomics End-to-end processing for RRBS, EM-seq and WGS; methylation calling, CpG filtering, and DMR detection; coverage/missingness evaluation and batch effect mitigation.

Advanced Statistics Linear Mixed Models (LMM/GLMM), quantitative genetics, and heritability estimation; high-dimensional data analysis and variance partitioning.

Workflow & HPC Pipeline orchestration using **Nextflow** (nf-core/methylseq); large-scale job scheduling via **SLURM** (job arrays) in HPC environments (LRZ, DSS).

Evolutionary Genomics Scalable variant/SNP calling pipelines; phylogenetics; population genomics workflows and genome-wide data management.

Experimental Design Power analysis for sequencing depth; cost-effective resequencing strategies; budget-constrained experimental planning and parameter benchmarking.

Method Development Custom repeatability metrics; locus reliability modeling; integration of genomic predictors; cross-platform data harmonization.

Data Management NCBI/SRA archival workflows; Aspera-based high-speed uploads; metadata curation and public repository management.

Molecular & Laboratory Skills

Molecular Techniques PCR, gel electrophoresis, nucleic acid extraction and cloning, bacterial transformation, agroinfiltration, CRISPR screening, cytogenetic karyotyping (G/C banding).

Teaching & Mentoring

2023–2025 **Graduate Teaching Assistant**, *Ludwig Maximilian University (LMU)*, Munich.
Assisted the *"Computational Biology with Python"* course. Led practical sessions on Python-based data analysis and managed course logistics via Moodle.

2020 **Guest Lecturer**, *University of West Attica*, Athens.
Delivered an invited lecture on *SARS-CoV-2 Genomics and Pathogenesis* for the undergraduate Virology curriculum.

2019 **Graduate Teaching Assistant**, *University of Crete*, Heraklion.
Instructional support for *BIO251: Functional Analysis of Biological Macromolecules* and *BIO208: Cellular and Genetic Analysis*. Demonstrated laboratory techniques and evaluated student performance in molecular assays.

2022–2023 **Lead Biology Instructor**, *"Thalis" Tutoring Center*, Athens.
Developed and delivered comprehensive biology curricula for senior secondary students; focused on preparing candidates for national-level examinations.

2014–2015 **Volunteer Science Tutor**, *Municipality of Patras*, Patras.
Provided biology and general science instruction at the Community Evening School, supporting adult learners and students from underprivileged backgrounds.

Languages Greek (native), English (C2), Spanish (B2), German (B1).

Referees

Prof. Jochen B. W. Wolf
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Personal webpage

Dr. Sarah Mueller
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